Banking-1A: Milestone 1 Data Exploration and Baseline Results

Online payment fraud detection on the IEEE-CIS dataset · Break Through Tech AI Studio with Mastercard · October 7, 2026

Summary

- We ran the full September pipeline on the real labeled data: **590,540 transactions, 3.50% fraud**, merged with the identity table (24% of transactions have identity data) and split by time into train, validation and a sealed blind test.
- Best validation result: **XGBoost with a small leakage-safe feature set (fe_v2), AUPRC 0.664**, up from 0.620 on raw features and ahead of LightGBM on raw features (0.640). Random guessing scores 0.034.
- If analysts review the riskiest 1% of transactions, 96% of what they review is fraud. Reviewing the top 5% catches 70% of all fraud and 67% of fraud dollars.
- Fraud is **not stable over time**: about 2.5% in the first month and 3.4% to 4.4% after. Twenty features shift a lot between month 1 and month 6. A random split would hide both effects.
- Whether a field is **missing is itself a strong signal**. Transactions with no billing address are 4.5 times as likely to be fraud; transactions with identity data are about 3.5 times as likely.
- The 1st-place Kaggle trick for rebuilding a customer ID **works on our data**, which makes October's behavioral features possible. It also shows that fraud labels spread across a customer's transactions.

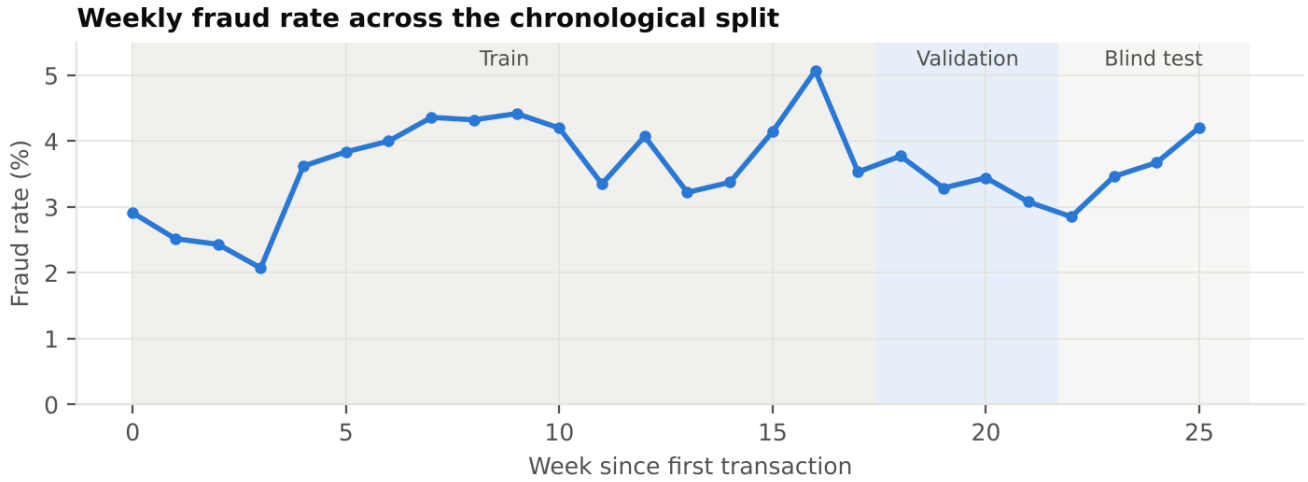
1. Data and split

The transaction and identity tables were joined on TransactionID with a left join, so every labeled transaction is kept, and a *has_identity* flag marks the 144,233 rows with identity data. The merged table has 435 columns and takes about 1 GB in memory after reading the anonymized numeric columns as 32-bit floats. The split is by day, so every model is tested on transactions that happen after the ones it learned from.

Split	Days	Transactions	Fraud rate	Use
Train	0 to 121	420,066	3.52%	Fit models and encoders
Validation	122 to 151	85,044	3.39%	Early stopping, model choice
Blind test	152 to 182	85,430	3.50%	Sealed until November

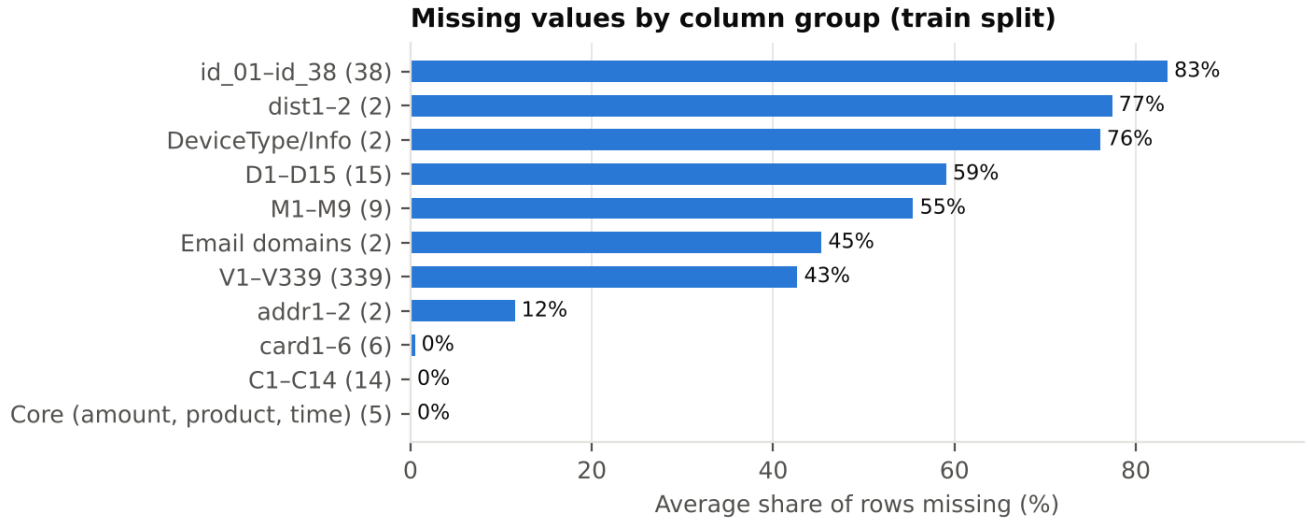
Split ID chrono-v1-c3eab22dc2 (a hash of the three frozen ID files). Every result below uses it.

2. Class imbalance and fraud over time (Task #16)

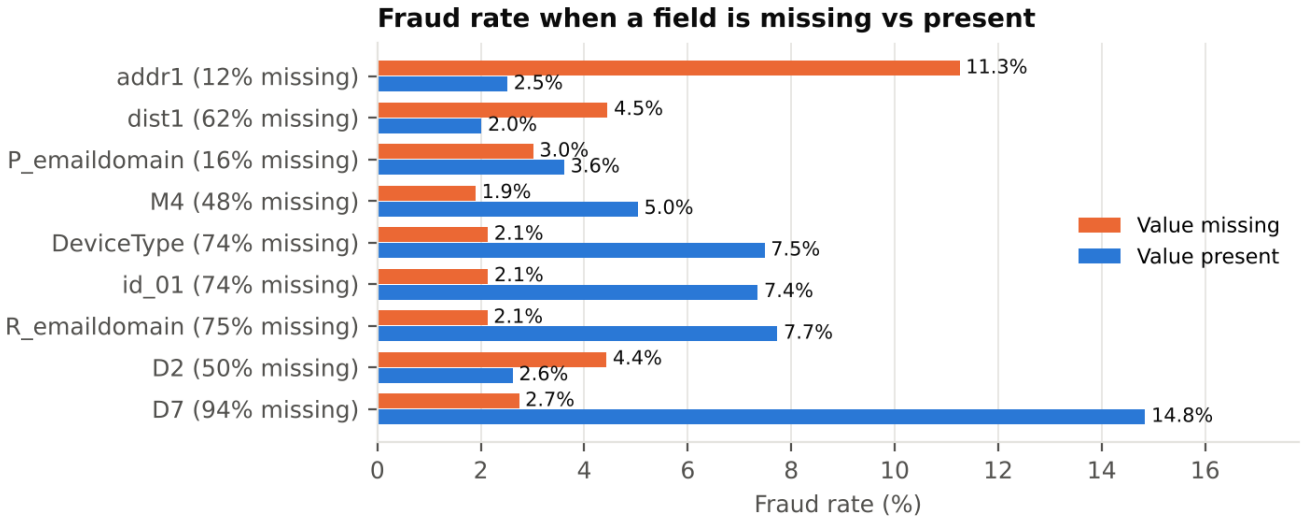


Only 1 in 29 transactions is fraud, so accuracy is useless: predicting "never fraud" scores 96.5%. We rank models by AUPRC and recall at a fixed review rate instead. The weekly fraud rate swings from about 2% to 5%. A chi-square test rejects a constant fraud rate across 30-day blocks ($p < 1e\text{-}100$). The validation window (3.39%) is a little calmer than training (3.52%), so validation precision may be slightly pessimistic.

3. Missing values (Task #8)

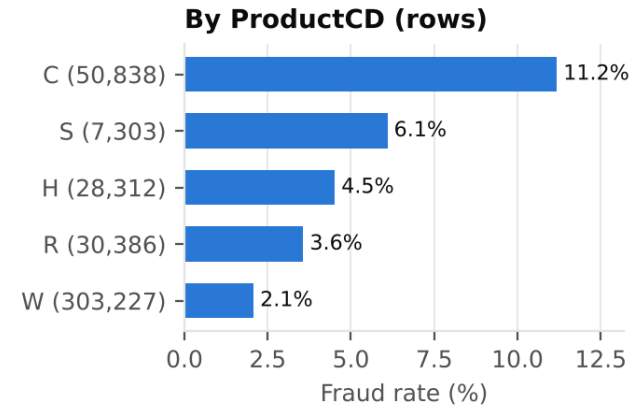


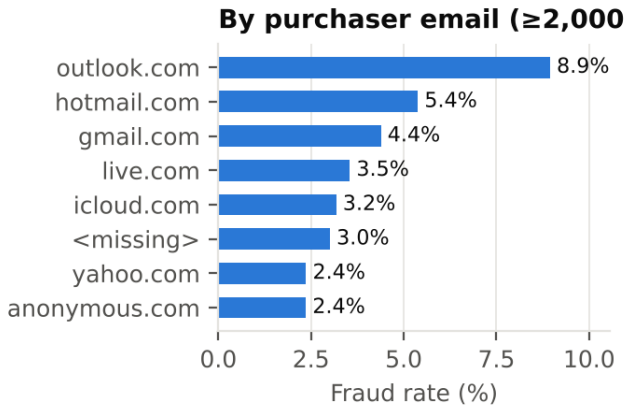
Missingness is heavy and comes in blocks. The identity columns (id_01 to id_38) are missing for 76% to 99% of transactions because only 24% have an identity record. The 339 V columns fall into 14 groups that are missing on exactly the same rows. That supports reducing V columns by group in October instead of treating them as 339 separate features. Only the C counts and core fields are complete.



Missing is not random. A missing billing address (addr1) goes with a 11.3% fraud rate vs 2.5% when present. Rows with identity, device or recipient-email data are about three times riskier, because those fields are mostly filled in for the riskier online channels. D7 is present on only 6% of rows, and those rows are 14.8% fraud. Takeaway: never fill these gaps silently. Keep a missing flag or use models that handle NaN directly, which all our tree models do.

4. Categorical fields (Task #10)





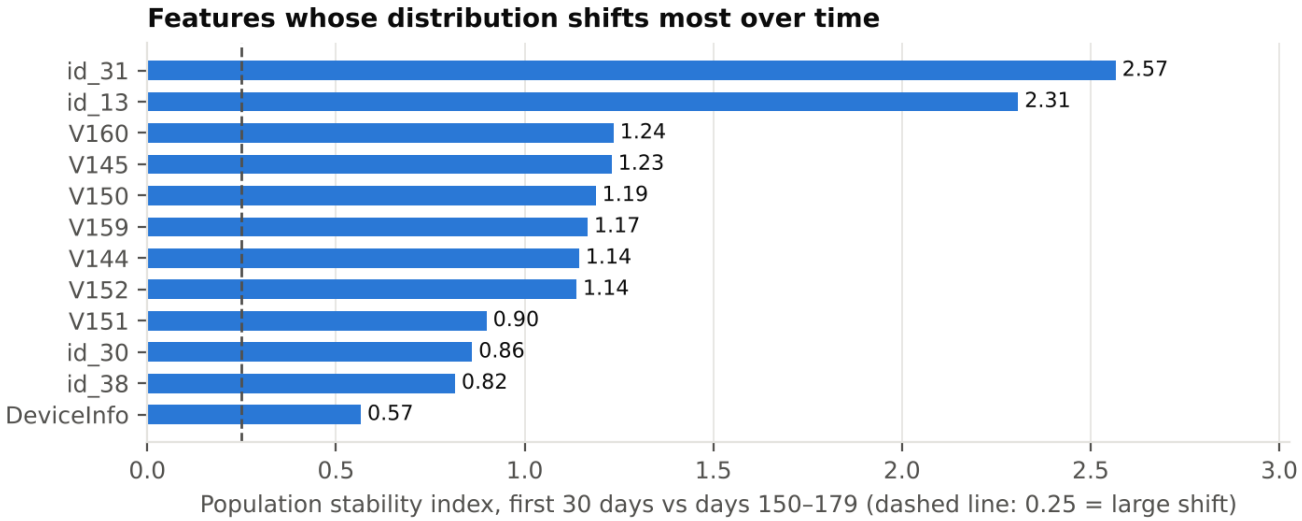
ProductCD is the clearest single risk signal: product C is 11.2% fraud, five times product W (2.1%), which is 72% of all transactions. Among common purchaser email domains, outlook.com (8.9%) and hotmail.com (5.4%) are well above gmail.com (4.4%) and yahoo.com (2.4%).

Field	Distinct values	Missing	Values with <50 rows	Encoding we suggest
card1	12,295	0%	11,337 (19% of rows)	Frequency / native categorical / embedding
DeviceInfo	1,556	78%	1,451 (11% of rows)	Group rare devices, then as above
card2	500	2%	53	Frequency or native categorical
addr1	318	12%	248	Frequency or native categorical
P_emaildomain	59	16%	6	One-hot after grouping rare domains
R_emaildomain	60	75%	27	One-hot after grouping rare domains

Field	Distinct values	Missing	Values with <50 rows	Encoding we suggest
ProductCD, card4, card6, M1–M9, DeviceType	2 to 5	0% to 74%	0 to 2	One-hot

card1 and DeviceInfo have thousands of rare values, so one-hot encoding them would add thousands of mostly empty columns. This matches the approach in Orator's CategoricalEncoder (frequency encoding for card1, grouping for DeviceInfo).

5. Feature drift



Twenty features have a population stability index above 0.25 between the first 30 days and days 150 to 179. The biggest are browser and OS versions (id_31, id_30), id_13, and a block of V columns (V144 to V166). Browser versions change naturally as people update, so a model that memorizes "Chrome 63" will age badly. These features are the first candidates for the time-consistency check in October.

6. Can we rebuild a customer ID? (Task #15)

Fraud labels in this dataset follow the customer: once a card has a chargeback, Vesta also labeled later transactions linked to that account, email or address as fraud. The data has no customer ID, but the 1st-place Kaggle team rebuilt one from card1, addr1 and D1 (days since the card was first seen). We tested three candidate keys. D3 (days since the previous transaction) is not used to build any key, so checking whether it matches the real gap between a key's transactions is an independent test.

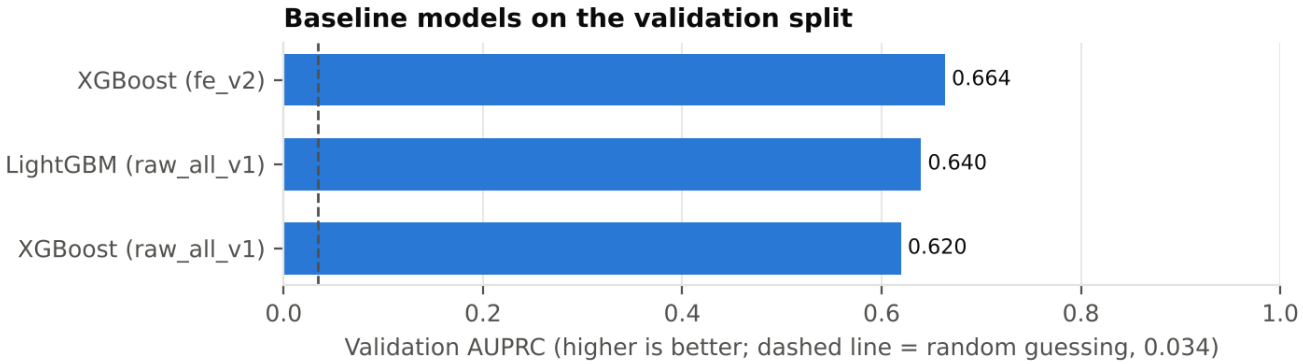
Key	Rows covered	Rows with repeat transactions	D3 matches the gap	Keys with one label only	Validation rows with training history
card1	100%	99%	36%	85%	99%
card1 + addr1	89%	97%	42%	90%	96%
card1 + addr1 + (day − D1)	89%	78%	**95%**	**98.5%**	50%

- The full key tracks one customer far better: D3 matches the real gap 95% of the time, against 42% for card1 + addr1.
- 98.5% of repeat keys carry a single label, and 55% of keys with any fraud are fraud on every transaction. This is the label spreading Vesta described. A model can score well by recognizing a known bad customer, so in November we should report results separately for first-time customers.

- Only half of validation transactions belong to a customer already seen in training, so October's behavioral features will be missing for the other half. Models must handle "no history".
- Decision: **go** for leakage-safe customer features in October (using only earlier transactions).

7. Baseline models (Tasks #18 and #19)

Every model used the same split, class weighting for the 1-in-29 imbalance (scale_pos_weight 27.4), early stopping on validation AUPRC, and the same metric code. *raw_all_v1* is all 432 raw columns (TransactionID and TransactionDT excluded). *fe_v2* adds a few leakage-safe features, described in section 8.



Model	Features	AUPRC	ROC-AUC	Fraud caught, top 1%	top 5%	KS	Train time
XGBoost	fe_v2	0.664	0.913	28.2%	69.8%	0.710	10 min
LightGBM	raw_all_v1	0.640	0.921	27.8%	67.7%	0.716	14 min
XGBoost	raw_all_v1	0.620	0.917	27.8%	65.8%	0.703	18 min

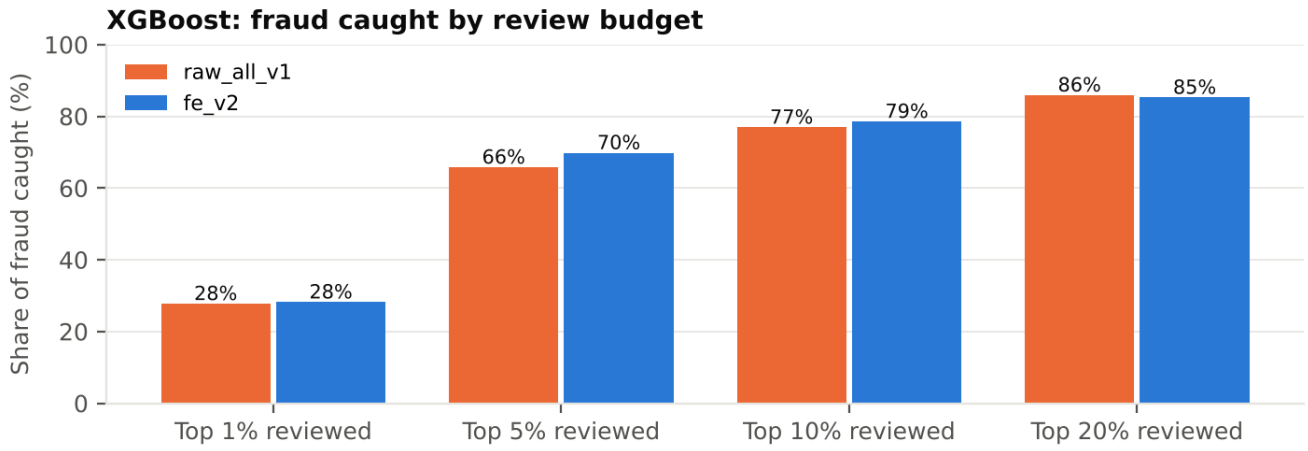
Note on the 1% column: the top 1% is 851 transactions and validation has 2,884 frauds, so the most any model can catch there is 29.5%. All three models are close to that ceiling, so the 5% column and AUPRC separate them better. Logistic regression, random forest and CatBoost (Tasks #17 and #20) haven't been run on the real data yet.

8. XGBoost baseline in detail (Task #19)

The raw XGBoost run fit the training data almost perfectly by round 1,000 while validation AUPRC crept up slowly and was still rising when it hit the 3,000-round cap, a sign of overfitting and drift. Its most important features were high-cardinality categories (DeviceInfo, email domains) and card1 treated as a plain number. The *fe_v2* feature set targets those problems. Every statistic is computed on the train split only, and none use the label:

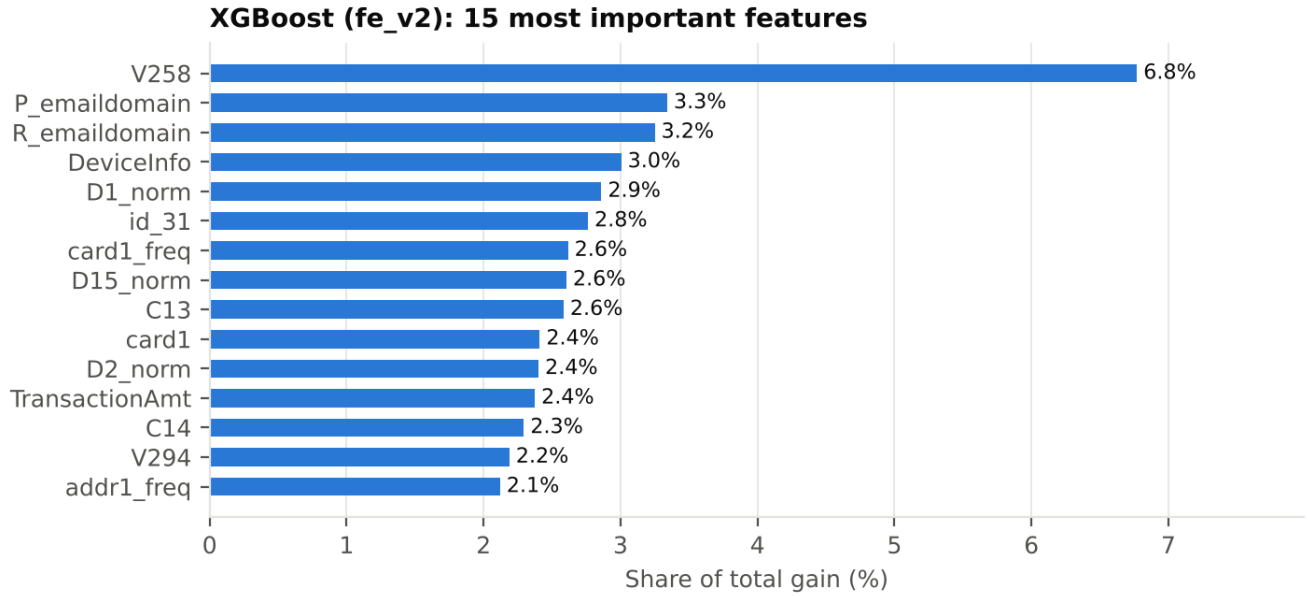
- **Frequency encodings** for card1, card2, card3, card5, addr1, addr2, both email domains, DeviceInfo and id_31 (values not seen in training get 0).
- **Normalized D columns:** D1, D2, D4, D10, D11 and D15 count days since a past event, so they grow with calendar time. They are replaced by day minus D, a fixed point in time.
- **Hour of day, cents of the amount, and log amount.**
- **max_cat_threshold = 32**, so a single split can't memorize long lists of rare categories.

Result: AUPRC 0.620 to **0.664 (+0.044)**, fraud caught in the top 5% 65.8% to 69.8%, and training stopped on its own at round 1,753 (10 minutes instead of 18).



Riskiest transactions reviewed (fe_v2)	Transactions	Fraud found	Share of fraud cases	Share of fraud dollars	Precision
Top 1%	851	814	28.2%	22.6%	95.7%
Top 5%	4,253	2,014	69.8%	67.2%	47.4%
Top 10%	8,505	2,266	78.6%	76.1%	26.6%
Top 20%	17,009	2,459	85.3%	84.1%	14.5%

At the 1% review rate our business framework assumes (about 28 reviews a day), 96% of reviewed transactions are fraud and only 0.05% of legitimate customers are flagged, far inside the 1% limit. The dollar share trails the case share in every bucket, so the model catches small frauds a little more easily than large ones. Amount-aware features are worth trying in October.



Three new features (D1_norm, card1_freq, D15_norm) are in the top 8. id_31 (browser version) is still high even though it drifts the most of any feature (section 5), so it should be grouped by browser family or dropped.

9. Limitations and next steps

- **One seed per run.** Our framework asks for three seeds before calling a winner. The +0.044 gain is well above the 0.01 bar, but it should be confirmed.
- **The four fe_v2 changes went in together**, so we don't yet know which one drives the gain. A cheap ablation for October.
- **Compute:** all runs used 2 CPU cores and 8 GB of memory, with learning rate 0.1 instead of 0.05. These are floors, not tuned results. Logistic regression, random forest and CatBoost still need real-data runs.
- **No customer-level features yet.** Section 6 shows they are feasible and probably the biggest remaining gain. They must use only earlier transactions.
- **Results are optimistic:** real chargebacks arrive weeks late, and labels spread across a customer's transactions. Cost and review-rate assumptions still need the Challenge Advisor's sign-off. The blind test split stays sealed until November.